

HDOC DAY-31

Name-Priyadarshna

Sapid-590032928

Batch-15

Question:

Write a C program to perform menu-driven operations on a given 2D array: find the sum of all elements, find the sum of each row and store it in an array, find the transpose of the matrix, and exit.

Aim:

To write a C program to perform menu-driven operations on a 2D array, such as finding the sum of all elements, row sums, and transpose of the matrix.

Algorithm

1. Start.
2. Enter the number of rows and columns.
3. Enter the matrix elements.
4. Display the menu.
5. Enter the user's choice.
6. If choice is **1**, calculate and display the sum of all elements.
7. If choice is **2**, calculate each row sum and store it in an array.
8. If choice is **3**, find and display the transpose of the matrix.
9. If choice is **4**, exit the program.
10. Otherwise, display **Invalid choice**.
11. Stop.

>C PROGRAM:

```
1 #include <stdio.h>
2
3 int main() {
4     int a[10][10]; sum[10];
5     int r, c, i, j, ch, total;
6
7     printf("Enter rows and columns: ");
8     scanf("%d %d", &r, &c);
9
10    printf("Enter matrix elements:");
11    for(i=0; i<r; i++)
12        for(j=0; j<c; j++)
13            scanf("%d", &a[i][j]);
14
15    do {
16        printf("1.Sum of all elements");
17        printf("2.Sum of each row");
18        printf("3.Transpose");
19        printf("\n4.Exit");
20        printf("Enter your choice: ");
21        scanf("%d", &ch);
22
23        switch(ch) {
24            case 1:
25                total=0;
26                for(i=0; i<r; i++)
27                    for(j=0; j<c; j++)
28                        total+=a[i][j];
29                printf("Sum of all elements = %d\n", total);
30                break;
31
32            case 2:
33                for(i=0; i<r; i++) {
34                    sum[i]=0;
35                    for(j=0; j<c; j++)
36                        sum[i]+=a[i][j];
37                    printf("Sum of row %d = %d\n", i+1, sum[i]);
38                }
39                break;
40
41            case 3:
42                printf("Transpose of matrix:\n");
43                for(j=0; j<c; j++) {
44                    for(i=0; i<r; i++)
45                        printf("%d ", a[i][j]);
46                    printf("\n");
47                }
48                break;
49
50            case 4:
51                printf("Exit\n");
52                break;
53            default:
54                printf("Invalid choice\n");
55        }
56    } while(ch>0);
57    return 0;
58 }
```

>SAMPLE EXECUTION:

Enter rows and columns: 2 3

Enter matrix elements:

1 2 3

4 5 6

1.Sum of all elements

2.Sum of each row

3.Transpose

Enter your choice: 1

Sum of all elements = 21

Enter your choice:2

Row Sums:

Sum of rows 1=6

Sum of rows 2=15

The row-sum array contains:[6, 15]

Enter the choice:3

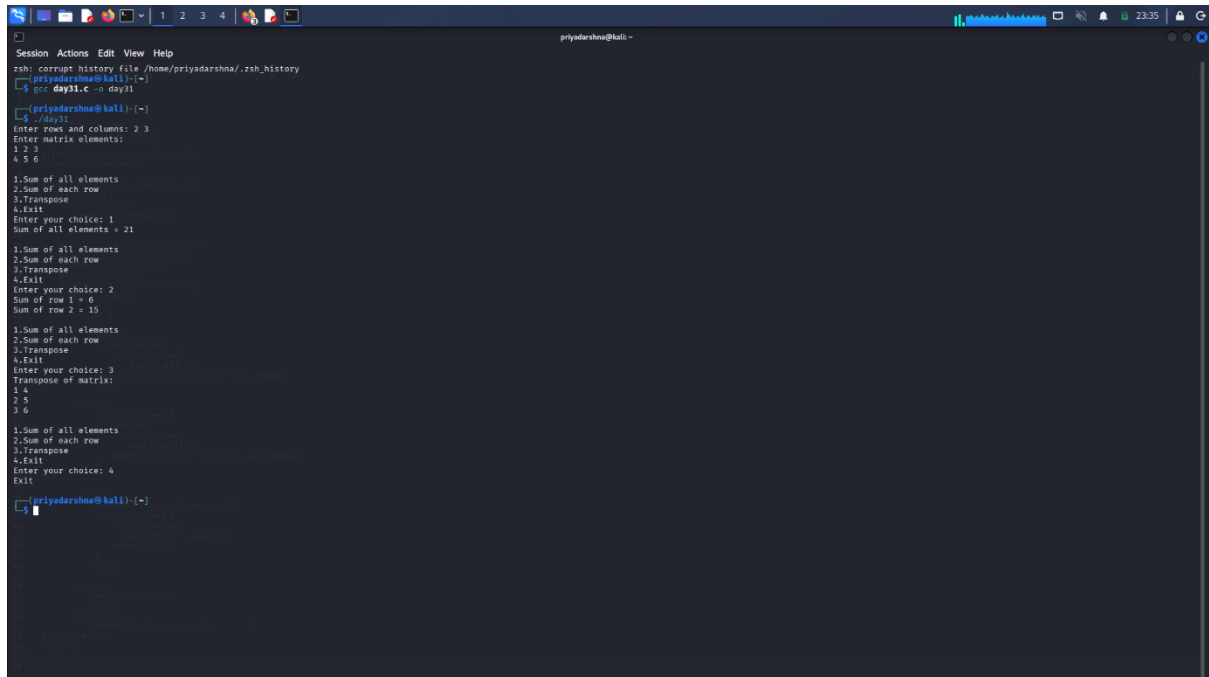
Transpose of the matrix:

1 4

2 5

3 6

>Execution:



```
Session Actions Edit View Help
prtyadarshna@kali: ~
zsh: corrupt history file /home/prtyadarshna/.zsh_history
prtyadarshna@kali: ~
$ gcc day31.c -o day31
prtyadarshna@kali: ~
$ ./day31
Enter rows and columns: 2 3
Enter matrix elements:
1 2 3
4 5 6
1.Sum of all elements
2.Sum of each row
3.Transpose
4.Exit
Enter your choice: 1
Sum of all elements = 21
1.Sum of all elements
2.Sum of each row
3.Transpose
4.Exit
Enter your choice: 2
Sum of row 1 = 6
Sum of row 2 = 15
1.Sum of all elements
2.Sum of each row
3.Transpose
4.Exit
Enter your choice: 3
Transpose of Matrix:
1 4
2 5
3 6
1.Sum of all elements
2.Sum of each row
3.Transpose
4.Exit
Enter your choice: 4
Exit
prtyadarshna@kali: ~
$
```